Importing libraries

```
import tensorflow as tf
from tensorflow import keras
import numpy as np
import matplotlib.pyplot as plt
import os
```

Loading datasets

```
(train_images, train_labels), (test_images, test_labels) =
tf.keras.datasets.cifar10.load_data()
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-
python.tar.gz
170500096/170498071 [==============================] - 4s 0us/step
170508288/170498071 [==============================] - 4s 0us/step
```

Data normalization

```
train_images = train_images / 255.0
test_images = test_images / 255.0
```

```
train_images.shape,test_images.shape
```

```
((50000, 32, 32, 3), (10000, 32, 32, 3))
```

```
train_labels = tf.keras.utils.to_categorical(train_labels)
test_labels = tf.keras.utils.to_categorical(test_labels)
```

```
train_images.shape,test_images.shape
```

```
((50000, 32, 32, 3), (10000, 32, 32, 3))
```

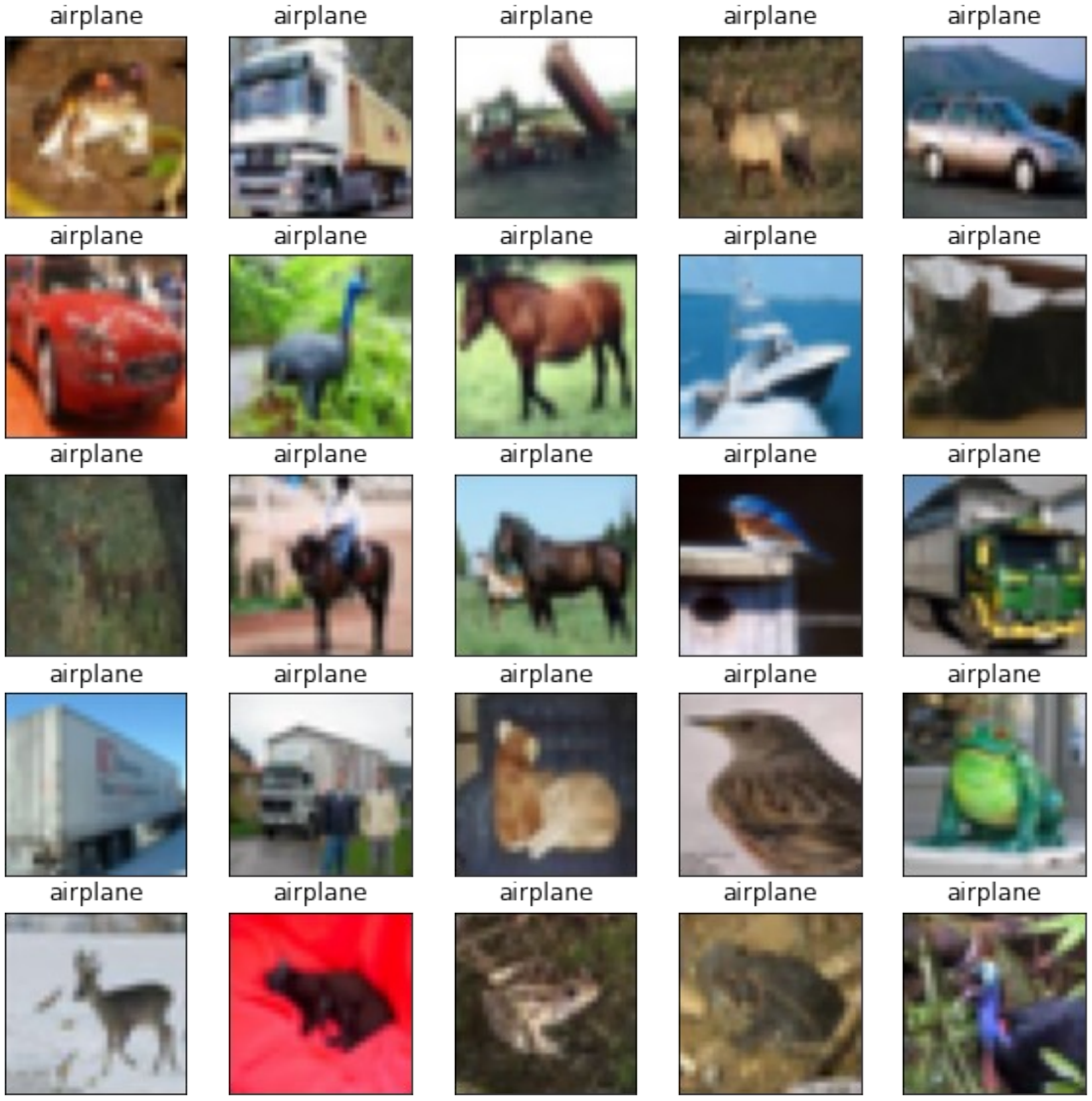
```
label_names = {
0 : 'airplane',
1 : 'automobile',

2 : 'bird',
3 : 'cat',
4 : 'deer',
5 : 'dog',
6 : 'frog',
7 : 'horse',
8 : 'ship',
9 : 'truck',
}
```

plotting images

```
rows, columns = 5, 5
fig = plt.figure(figsize=(10,10))
for i in range(1,columns*rows+1):
```

```
  fig.add_subplot(rows, columns, i)
  plt.imshow(train_images[i-1])
  plt.xticks([])
  plt.yticks([])
  plt.title("{}".format(label_names[train_labels[i-1][0]]))
plt.show();
```



Callbacks

```
filepath = '/content/drive/MyDrive/Colab Notebooks/week7.h5'
checkpoint = tf.keras.callbacks.ModelCheckpoint(filepath,monitor =
'val_loss', verbose = 1, save_best_only = True, mode = 'min')

stop_early = tf.keras.callbacks.EarlyStopping(monitor
='val_accuracy',mode = 'auto', patience = 5)
reduce_lr =
tf.keras.callbacks.ReduceLROnPlateau(monitor='val_loss',factor = 0.2,
```

```
patience = 5,mode = 'auto', min_lr = 0.0001)
callbacklist = [stop_early,reduce_lr,checkpoint]

from tensorflow.keras.regularizers import l1,l2
weight_decay = 1e-4
batch_size = 64
initial_epochs = 10
lr = 0.001
lr_decay = 1e-6
def model_build():
  inputs = keras.Input(shape=(32, 32, 3))
  x = tf.keras.layers.Conv2D(filters = 32,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer = l2(weight_decay)
                             )(inputs)
  x = tf.keras.layers.BatchNormalization()(x)

  x = tf.keras.layers.MaxPool2D(pool_size=(2,2))(x)
  x = tf.keras.layers.Dropout(rate = 0.2)(x)

  x = tf.keras.layers.Conv2D(filters = 64,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer = l2(weight_decay)
                             )(x)
  x = tf.keras.layers.BatchNormalization()(x)


  x = tf.keras.layers.MaxPool2D(pool_size=(2,2))(x)

  x = tf.keras.layers.Dropout(rate = 0.3)(x)

  x = tf.keras.layers.Conv2D(filters = 128,
                             kernel_size = (3,3),
                             padding = 'same',
                             activation = 'relu',
                             kernel_regularizer =  l2(weight_decay)
                             )(x)
  x = tf.keras.layers.BatchNormalization()(x)


  x = tf.keras.layers.MaxPool2D(pool_size=(2,2))(x)

  x = tf.keras.layers.Dropout(rate = 0.4)(x)

  x = tf.keras.layers.Flatten()(x)
```

```
  x = tf.keras.layers.Dense(1024, activation = 'relu')(x)
  x = tf.keras.layers.Dropout(rate = 0.3)(x)

  x = tf.keras.layers.Dense(256, activation = 'relu')(x)
  x = tf.keras.layers.Dropout(rate = 0.2)(x)

  outputs = tf.keras.layers.Dense(10, activation = 'softmax')(x)
  model = tf.keras.Model(inputs = inputs,outputs=outputs, name
='cifar-10')

  model.compile(loss = tf.keras.losses.CategoricalCrossentropy(),
                optimizer = tf.keras.optimizers.Adam(learning_rate=
lr, decay = lr_decay),
                metrics=['accuracy'])
  return model

my_model = model_build()
my_model.summary()
```

```
Model: "cifar-10"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 input_1 (InputLayer)        [(None, 32, 32, 3)]       0

 conv2d (Conv2D)             (None, 32, 32, 32)        896

 batch_normalization (BatchN  (None, 32, 32, 32)       128
 ormalization)

 max_pooling2d (MaxPooling2D  (None, 16, 16, 32)       0
 )

 dropout (Dropout)           (None, 16, 16, 32)        0

 conv2d_1 (Conv2D)           (None, 16, 16, 64)        18496

 batch_normalization_1 (Batc  (None, 16, 16, 64)       256
 hNormalization)

 max_pooling2d_1 (MaxPooling  (None, 8, 8, 64)         0
 2D)

 dropout_1 (Dropout)         (None, 8, 8, 64)          0

 conv2d_2 (Conv2D)           (None, 8, 8, 128)         73856

 batch_normalization_2 (Batc  (None, 8, 8, 128)        512
 hNormalization)
```

```
 max_pooling2d_2 (MaxPooling  (None, 4, 4, 128)        0
 2D)

 dropout_2 (Dropout)         (None, 4, 4, 128)         0

 flatten (Flatten)           (None, 2048)              0

 dense (Dense)               (None, 1024)              2098176

 dropout_3 (Dropout)         (None, 1024)              0

 dense_1 (Dense)             (None, 256)               262400

 dropout_4 (Dropout)         (None, 256)               0

 dense_2 (Dense)             (None, 10)                2570

=================================================================
Total params: 2,457,290
Trainable params: 2,456,842
Non-trainable params: 448
_________________________________________________________________
```

```
epochs = 10
batch_size = 64
h1 = my_model.fit(train_images,
                  train_labels,
                  validation_split=0.2,
                  epochs=epochs,

                  callbacks = callbacklist)
```

```
Epoch 1/10
1250/1250 [==============================] - ETA: 0s - loss: 1.6592 -
accuracy: 0.4160
Epoch 1: val_loss improved from inf to 1.27934, saving model to
/content/drive/MyDrive/Colab Notebooks/week7.h5
1250/1250 [==============================] - 31s 17ms/step - loss:
1.6592 - accuracy: 0.4160 - val_loss: 1.2793 - val_accuracy: 0.5526 -
lr: 0.0010
Epoch 2/10
1250/1250 [==============================] - ETA: 0s - loss: 1.2943 -
accuracy: 0.5482
Epoch 2: val_loss did not improve from 1.27934
1250/1250 [==============================] - 21s 17ms/step - loss:
1.2943 - accuracy: 0.5482 - val_loss: 1.4340 - val_accuracy: 0.5060 -
lr: 0.0010
Epoch 3/10
1248/1250 [============================>.] - ETA: 0s - loss: 1.1400 -
accuracy: 0.6090
```

```
Epoch 3: val_loss improved from 1.27934 to 1.10018, saving model to
/content/drive/MyDrive/Colab Notebooks/week7.h5
1250/1250 [==============================] - 21s 17ms/step - loss:
1.1400 - accuracy: 0.6089 - val_loss: 1.1002 - val_accuracy: 0.6242 -
lr: 0.0010
Epoch 4/10
1249/1250 [============================>.] - ETA: 0s - loss: 1.0479 -
accuracy: 0.6511
Epoch 4: val_loss did not improve from 1.10018
1250/1250 [==============================] - 20s 16ms/step - loss:
1.0478 - accuracy: 0.6511 - val_loss: 2.1687 - val_accuracy: 0.4496 -
lr: 0.0010
Epoch 5/10
1250/1250 [==============================] - ETA: 0s - loss: 0.9809 -
accuracy: 0.6759
Epoch 5: val_loss improved from 1.10018 to 0.87004, saving model to
/content/drive/MyDrive/Colab Notebooks/week7.h5
1250/1250 [==============================] - 21s 16ms/step - loss:
0.9809 - accuracy: 0.6759 - val_loss: 0.8700 - val_accuracy: 0.7146 -
lr: 0.0010
Epoch 6/10
1247/1250 [============================>.] - ETA: 0s - loss: 0.9339 -
accuracy: 0.6951
Epoch 6: val_loss did not improve from 0.87004
1250/1250 [==============================] - 21s 17ms/step - loss:
0.9341 - accuracy: 0.6949 - val_loss: 1.6330 - val_accuracy: 0.5474 -
lr: 0.0010
Epoch 7/10
1249/1250 [============================>.] - ETA: 0s - loss: 0.8877 -
accuracy: 0.7108
Epoch 7: val_loss did not improve from 0.87004
1250/1250 [==============================] - 20s 16ms/step - loss:
0.8877 - accuracy: 0.7108 - val_loss: 0.9863 - val_accuracy: 0.6741 -
lr: 0.0010
Epoch 8/10
1250/1250 [==============================] - ETA: 0s - loss: 0.8551 -
accuracy: 0.7279
Epoch 8: val_loss did not improve from 0.87004
1250/1250 [==============================] - 20s 16ms/step - loss:
0.8551 - accuracy: 0.7279 - val_loss: 0.9027 - val_accuracy: 0.7182 -
lr: 0.0010
Epoch 9/10
1247/1250 [============================>.] - ETA: 0s - loss: 0.8218 -
accuracy: 0.7414
Epoch 9: val_loss improved from 0.87004 to 0.79528, saving model to
/content/drive/MyDrive/Colab Notebooks/week7.h5
1250/1250 [==============================] - 21s 17ms/step - loss:
0.8218 - accuracy: 0.7413 - val_loss: 0.7953 - val_accuracy: 0.7527 -
lr: 0.0010
Epoch 10/10
```

```
1249/1250 [============================>.] - ETA: 0s - loss: 0.8047 -
accuracy: 0.7473
Epoch 10: val_loss did not improve from 0.79528
1250/1250 [==============================] - 20s 16ms/step - loss:
0.8048 - accuracy: 0.7473 - val_loss: 0.8963 - val_accuracy: 0.7269 -
lr: 0.0010
```

```
total_epochs = epochs + 30
h2 = my_model.fit(train_images,train_labels,
                  validation_split = 0.2,
                  initial_epoch = h1.epoch[-1],
                  epochs = total_epochs,
                  callbacks = callbacklist)
```

```
Epoch 10/40
1249/1250 [============================>.] - ETA: 0s - loss: 0.7763 -
accuracy: 0.7571
Epoch 10: val_loss did not improve from 0.79528
1250/1250 [==============================] - 19s 15ms/step - loss:
0.7763 - accuracy: 0.7571 - val_loss: 0.8108 - val_accuracy: 0.7454 -
lr: 0.0010
Epoch 11/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.7612 -
accuracy: 0.7648
Epoch 11: val_loss improved from 0.79528 to 0.76773, saving model
to /content/drive/MyDrive/Colab Notebooks/week7.h5
1250/1250 [==============================] - 19s 15ms/step - loss:
0.7608 - accuracy: 0.7649 - val_loss: 0.7677 - val_accuracy: 0.7719 -
lr: 0.0010
Epoch 12/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.7361 -
accuracy: 0.7740
Epoch 12: val_loss did not improve from 0.76773
1250/1250 [==============================] - 19s 15ms/step - loss:
0.7362 - accuracy: 0.7739 - val_loss: 0.7872 - val_accuracy: 0.7664 -
lr: 0.0010
Epoch 13/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.7140 -
accuracy: 0.7830
Epoch 13: val_loss did not improve from 0.76773
1250/1250 [==============================] - 19s 15ms/step - loss:
0.7139 - accuracy: 0.7829 - val_loss: 0.9432 - val_accuracy: 0.7205 -
lr: 0.0010
Epoch 14/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.6938 -
accuracy: 0.7911
Epoch 14: val_loss did not improve from 0.76773
1250/1250 [==============================] - 19s 15ms/step - loss:
0.6940 - accuracy: 0.7911 - val_loss: 1.1209 - val_accuracy: 0.6985 -
lr: 0.0010
Epoch 15/40
```

```
1248/1250 [============================>.] - ETA: 0s - loss: 0.6767 -
accuracy: 0.7974
Epoch 15: val_loss did not improve from 0.76773
1250/1250 [==============================] - 19s 15ms/step - loss:
0.6769 - accuracy: 0.7973 - val_loss: 1.2383 - val_accuracy: 0.6678 -
lr: 0.0010
Epoch 16/40
1250/1250 [==============================] - ETA: 0s - loss: 0.6583 -
accuracy: 0.8011
Epoch 16: val_loss improved from 0.76773 to 0.73617, saving model
to /content/drive/MyDrive/Colab Notebooks/week7.h5
1250/1250 [==============================] - 21s 17ms/step - loss:
0.6583 - accuracy: 0.8011 - val_loss: 0.7362 - val_accuracy: 0.7901 -
lr: 0.0010
Epoch 17/40
1250/1250 [==============================] - ETA: 0s - loss: 0.6512 -
accuracy: 0.8095
Epoch 17: val_loss did not improve from 0.73617
1250/1250 [==============================] - 20s 16ms/step - loss:
0.6512 - accuracy: 0.8095 - val_loss: 0.7654 - val_accuracy: 0.7813 -
lr: 0.0010
Epoch 18/40
1247/1250 [============================>.] - ETA: 0s - loss: 0.6290 -
accuracy: 0.8136
Epoch 18: val_loss did not improve from 0.73617
1250/1250 [==============================] - 23s 18ms/step - loss:
0.6291 - accuracy: 0.8135 - val_loss: 0.7484 - val_accuracy: 0.7859 -
lr: 0.0010
Epoch 19/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.6195 -
accuracy: 0.8186
Epoch 19: val_loss did not improve from 0.73617
1250/1250 [==============================] - 23s 19ms/step - loss:
0.6193 - accuracy: 0.8187 - val_loss: 0.7744 - val_accuracy: 0.7855 -
lr: 0.0010
Epoch 20/40
1250/1250 [==============================] - ETA: 0s - loss: 0.6061 -
accuracy: 0.8255
Epoch 20: val_loss improved from 0.73617 to 0.73229, saving model
to /content/drive/MyDrive/Colab Notebooks/week7.h5
1250/1250 [==============================] - 22s 18ms/step - loss:
0.6061 - accuracy: 0.8255 - val_loss: 0.7323 - val_accuracy: 0.7967 -
lr: 0.0010
Epoch 21/40
1250/1250 [==============================] - ETA: 0s - loss: 0.5901 -
accuracy: 0.8282
Epoch 21: val_loss did not improve from 0.73229
1250/1250 [==============================] - 22s 18ms/step - loss:
0.5901 - accuracy: 0.8282 - val_loss: 1.1178 - val_accuracy: 0.7116 -
lr: 0.0010
```

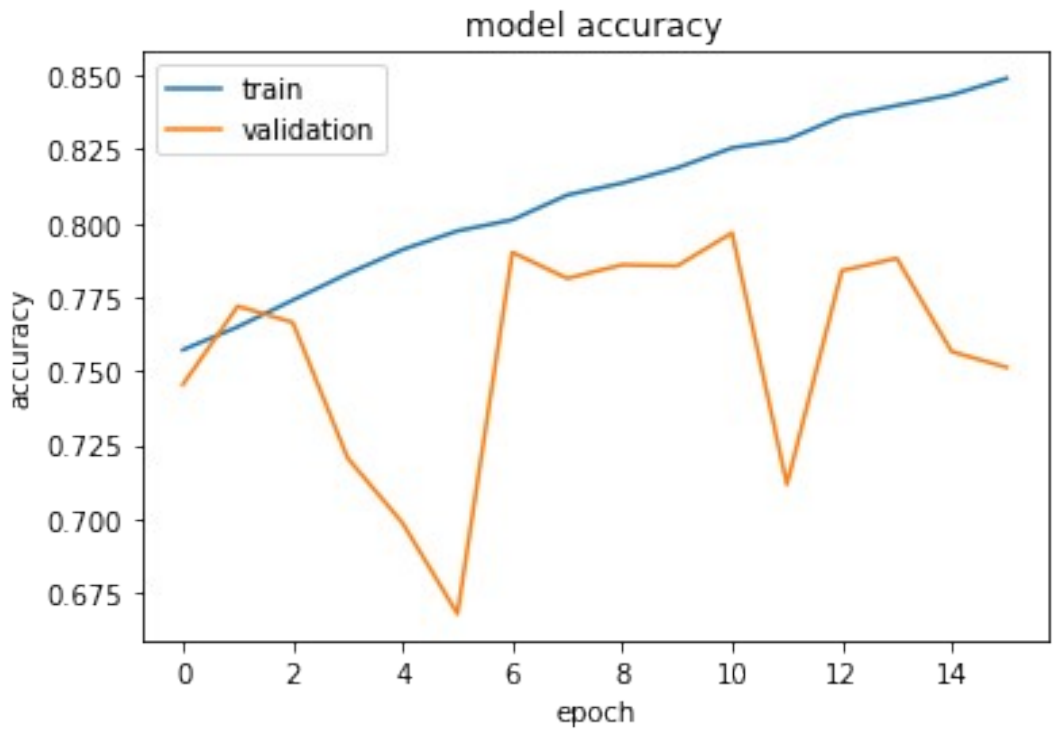
```
Epoch 22/40
1247/1250 [============================>.] - ETA: 0s - loss: 0.5785 -
accuracy: 0.8361
Epoch 22: val_loss did not improve from 0.73229
1250/1250 [==============================] - 23s 18ms/step - loss:
0.5782 - accuracy: 0.8361 - val_loss: 0.7859 - val_accuracy: 0.7839 -
lr: 0.0010
Epoch 23/40
1250/1250 [==============================] - ETA: 0s - loss: 0.5643 -
accuracy: 0.8398
Epoch 23: val_loss did not improve from 0.73229
1250/1250 [==============================] - 20s 16ms/step - loss:
0.5643 - accuracy: 0.8398 - val_loss: 0.7736 - val_accuracy: 0.7881 -
lr: 0.0010
Epoch 24/40
1248/1250 [============================>.] - ETA: 0s - loss: 0.5489 -
accuracy: 0.8435
Epoch 24: val_loss did not improve from 0.73229
1250/1250 [==============================] - 21s 17ms/step - loss:
0.5491 - accuracy: 0.8434 - val_loss: 0.9498 - val_accuracy: 0.7565 -
lr: 0.0010
Epoch 25/40
1249/1250 [============================>.] - ETA: 0s - loss: 0.5373 -
accuracy: 0.8490
Epoch 25: val_loss did not improve from 0.73229
1250/1250 [==============================] - 21s 17ms/step - loss:
0.5372 - accuracy: 0.8490 - val_loss: 0.9288 - val_accuracy: 0.7512 -
lr: 0.0010
```

Performance

Accuracy

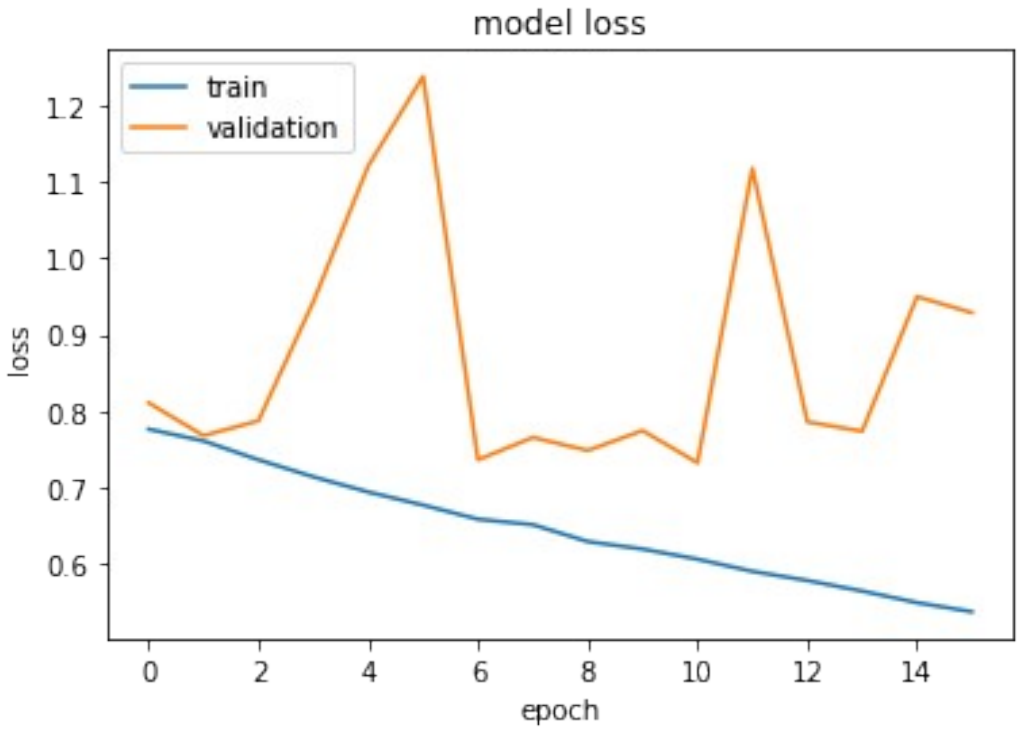
```
print(h2.history.keys)
plt.plot(h2.history['accuracy'])
plt.plot(h2.history['val_accuracy'])
plt.title('model accuracy')
plt.ylabel('accuracy')
plt.xlabel('epoch')
plt.legend(['train', 'validation'], loc='upper left')
plt.show()
```

```
<built-in method keys of dict object at 0x7fe1c02af820>
```



Loss

```
plt.plot(h2.history['loss'])
plt.plot(h2.history['val_loss'])
plt.title('model loss')
plt.ylabel('loss')
plt.xlabel('epoch')
plt.legend(['train', 'validation'], loc='upper left')
plt.show()
```



Evaluation

```
test_loss, test_acc = my_model.evaluate(test_images, test_labels)
print('\nTest accuracy: {}'.format(test_acc))
```

```
313/313 [==============================] - 2s 6ms/step - loss: 0.9604
- accuracy: 0.7444

Test accuracy: 0.7444000244140625
```